

Their research will allow fully-assembled electric and electronic gadgets to be printed in one go, instead of going the traditional way—creating a casing and then filling it with electronic circuit boards and components. Printing a complete and fully-assembled device by printing layer upon layer of conducting and semiconducting polymers, the device circuitry is built up as part of the bodywork.

When the technique is perfected, devices such as light bulbs, cell phones, radios, remote controls and toys will be spat out from printers as individual, fully-functional systems without expensive and labour-intensive production on an assembly line.

Three-dimensional printers are valuable tools for making prototypes of new designs. They deposit layers made from droplets of smart polymers, which gradually build up into 3D shapes. (Smart polymers are fascinating materials that show distinct responses to differences and variations in the environment, such as thermal gradient. They are called 'smart' because one can actually harvest their odd properties to benefit a wide range of applications.)

Such printing techniques have now become so sophisticated that it is possible to print working prototypes with mechanical parts that move just as they would in the final product.

The University of Berkeley's important addition to this art is to allow the electronics to be included in the printed device, rather than being added at a cost later.

Already, the Berkeley team has worked out how to print out electronic components such as transistors, capacitors, inductive coils and other semiconductor components. "These may be connected to form complete circuits for actuation and control," says Canny. He also believes that once they develop inkjet cartridges that can handle all the polymers needed for casing and circuit printing, they would be able to make, say, a remote control.

It would be printed as a single component, rather than being built up of smaller components. The infrared emitter and all the other electronics would be polymer-based.

Is There Anything We Can't Print?

Brains, probably!

There are so many uses for the inkjet, 3D or otherwise, that it's overwhelming. You sort of get the idea that there is nothing that cannot be printed out.

The point here is twofold: first, desktop manufacturing will revolutionise the way we think of products. With flexonics maturing, and with high-quality 3D printers getting cheaper, we will soon enter an age where, like we mentioned, you will pay for plans rather than for parts.

Second, one needs to remember that not all materials and substrates have been researched. For example, to print out, say, a handbag, you would need a process that could extrude leather out of a nozzle, which is not very realistic, at least for now.

Another limiting factor is how easily plans can be laid out for the object: if it's something like a screwdriver, it's relatively easy because the dimensions and other properties are well-known and easily documented. And as we mentioned, it would be difficult to print out a handbag because it's not quite so easily defined.

Desktop manufacturing could easily become a buzzword in the near future. Remember, you read about it first in *Digit!* Research in the areas we've mentioned is ongoing and exciting. Roger Spielman of Boeing probably sums it up best: "It's actually happening. You're going to hear a lot about this in the near future. There are hundreds of uses. You can build a part in a part. You can build a ship in a bottle. We can make real things out of dust." ☒

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1/2 pg AD